

Aim: Determination of Reynold's Number for laminar, turbulent and transition flow.

Objective: To study the laminar, turbulent and transition flow and to determine the Reynold's number.

Apparatus Used: Reynold's Apparatus

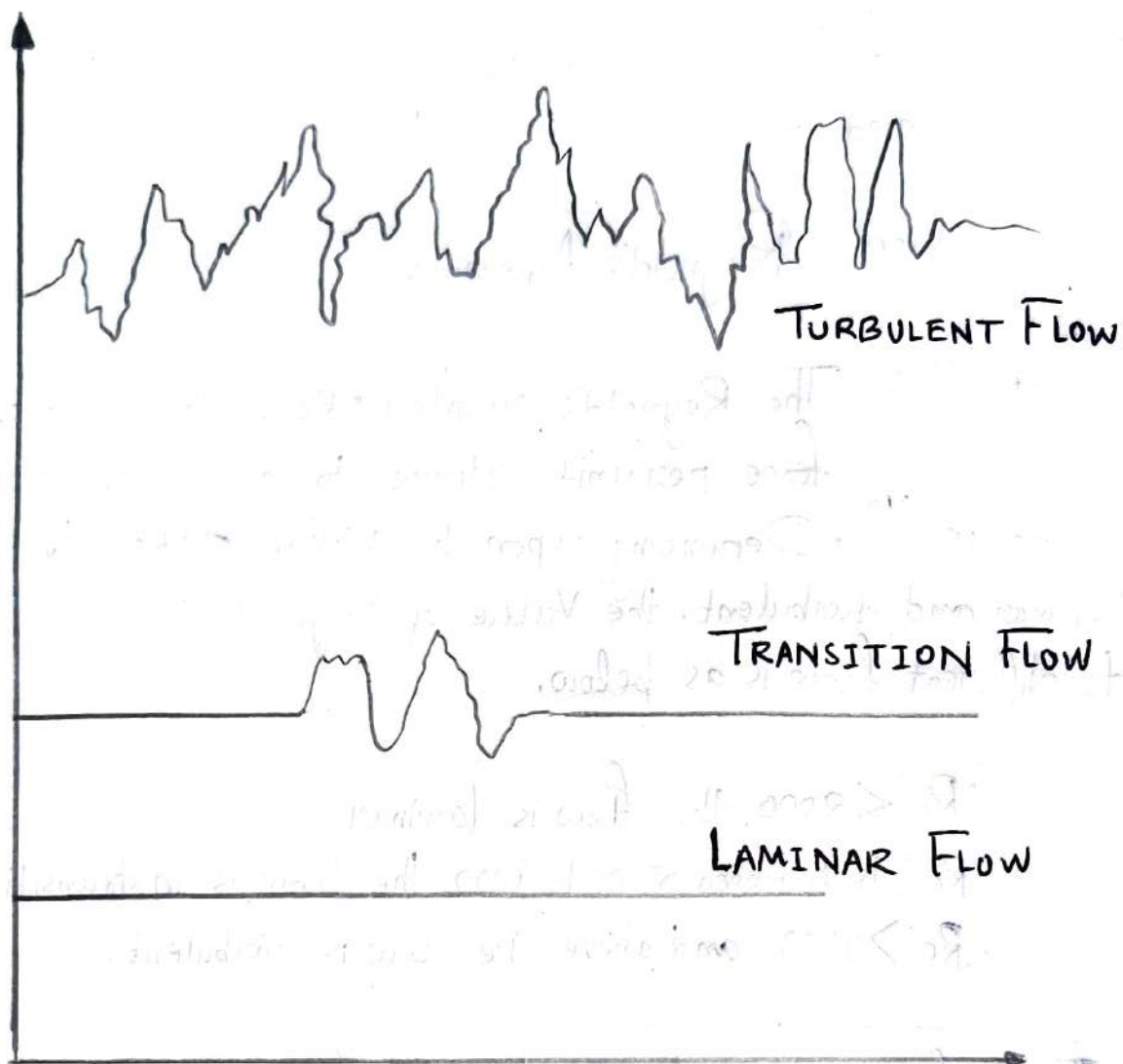
Introduction: The Reynold's number 'Re' is the ratio of inertial force per unit Volume to the viscous force per unit Volume i.e. $\frac{\rho V D}{\mu}$. Depending upon the value of 'Re' flow can be laminar and turbulent. The value of Reynold's Number corresponding to different flow is as below,

'Re' < 2000 the flow is laminar

'Re' is between 2000 to 4000 the flow is in transition stage.

'Re' > 4000 and above the flow is turbulent.

The Velocity at which the laminar flow changes to turbulent flow is called the upper Critical Velocity.



DYE STREAKS FOR THE THESE TYPES OF FLOW

Description of the Apparatus:

The apparatus consists of a hydraulic bench where a reservoir is used. A motor of 0.5 HP (Centrifugal pump) is used for water circulation within the bench. Initially, the water is pumped into the inlet tank and the flow is controlled with the help of Valve (ball type) attached to the bench. A discharge or volumetric tank, attached with a piezometer tube is used to measure the rate of discharge of the flow. A dye container is attached to the bench for understanding the type of flow. A glass tube having a bell mouthed entrance at one end and a regulating Valve at the other is provided. After some time the bench is ready to perform with the experiment.

Experimental Procedure:

1. Fill the tank with water.
2. Open the regulating Valve slowly so that the jet of dye flows uniformly through the tube i.e. laminar flow.
3. Calculate the discharge and also Calculate the Reynold's number.
4. Increase the flow of water slowly to calculate the Reynold's Number at the transition state and the turbulent flow.
5. Also see the mode of dye in the tube.

Observation Table :

SL No.	Initial level, h_1 (cm)	Final level, h_2 (cm)	Time, t (Sec)	Discharge, $Q = \frac{(h_2 - h_1)}{L} \times A$ (cm ³ /sec)
5				
1.	2.5	2.8	10	$Q = \frac{(2.8 - 2.5)}{10} \times 750 = 22.5$
2.	2.8	3	10	$Q = \frac{(3 - 2.8)}{10} \times 750 = 15$
3.	3	3.2	10	$Q = \frac{(3.2 - 3)}{10} \times 750 = 17.5$
10				$\text{Avg.} = \frac{52.5}{3} = 17.5$
1.	2	2.8	10	$Q = \frac{(2.8 - 2)}{10} \times 750 = 60$
2.	2.8	3.6	10	$Q = \frac{(3.6 - 2.8)}{10} \times 750 = 60$
15	3.	3.6	10	$Q = \frac{(3.6 - 3)}{10} \times 750 = 45$
				$\text{Avg.} = \frac{135}{3} = 45$
1.	2.2	4.6	10	$Q = \frac{(4.6 - 2.2)}{10} \times 750 = 180$
2.	4.6	5.8	10	$Q = \frac{(5.8 - 4.6)}{10} \times 750 = 90$
20	3.	5.8	10	$Q = \frac{(5.8 - 5.8)}{10} \times 750 = 82.5$
				$\text{Avg.} = \frac{352.5}{3} = 117.5$

25

Result :- Laminar flow :
 Turbulent flow :
 Transition flow :

Aim:- To determine the output power of pelton turbine
To determine the efficiency of the pelton turbine.

Objective: To study the operation of pelton wheel turbine.

Introduction: The pelton wheel is a tangential flow impulse turbine. The water strikes the bucket along the tangent of the runner. The energy available at the inlet of the turbine is only kinematic energy. The pressure at the inlet and outlet of the turbine is atmosphere. The turbine is used for high head.

PELTON WHEEL TURBINE TEST RIG:

The pelton wheel turbine as shown in fig 3.3 Consist of Casing with a large Circular transparent window kept at the front for the visual inspection of the jet on buckets a bearing pedestals rotor assembly of shaft, runner and brake drum, all mounted on Suitable Sturdy Cast iron base plates it. pelton wheel is a tangential flow impulse turbine. The water strikes the bucket along the tangent of the runner. The energy available at the inlet of the turbine is only Kinetic energy. The pressure at the inlet and outlet of the turbine is atmosphere. The turbine is used for high head. In an impulse turbine, all the available energy of water or velocity head by passing it through a Contracting nozzle provided at the end of the penstock.

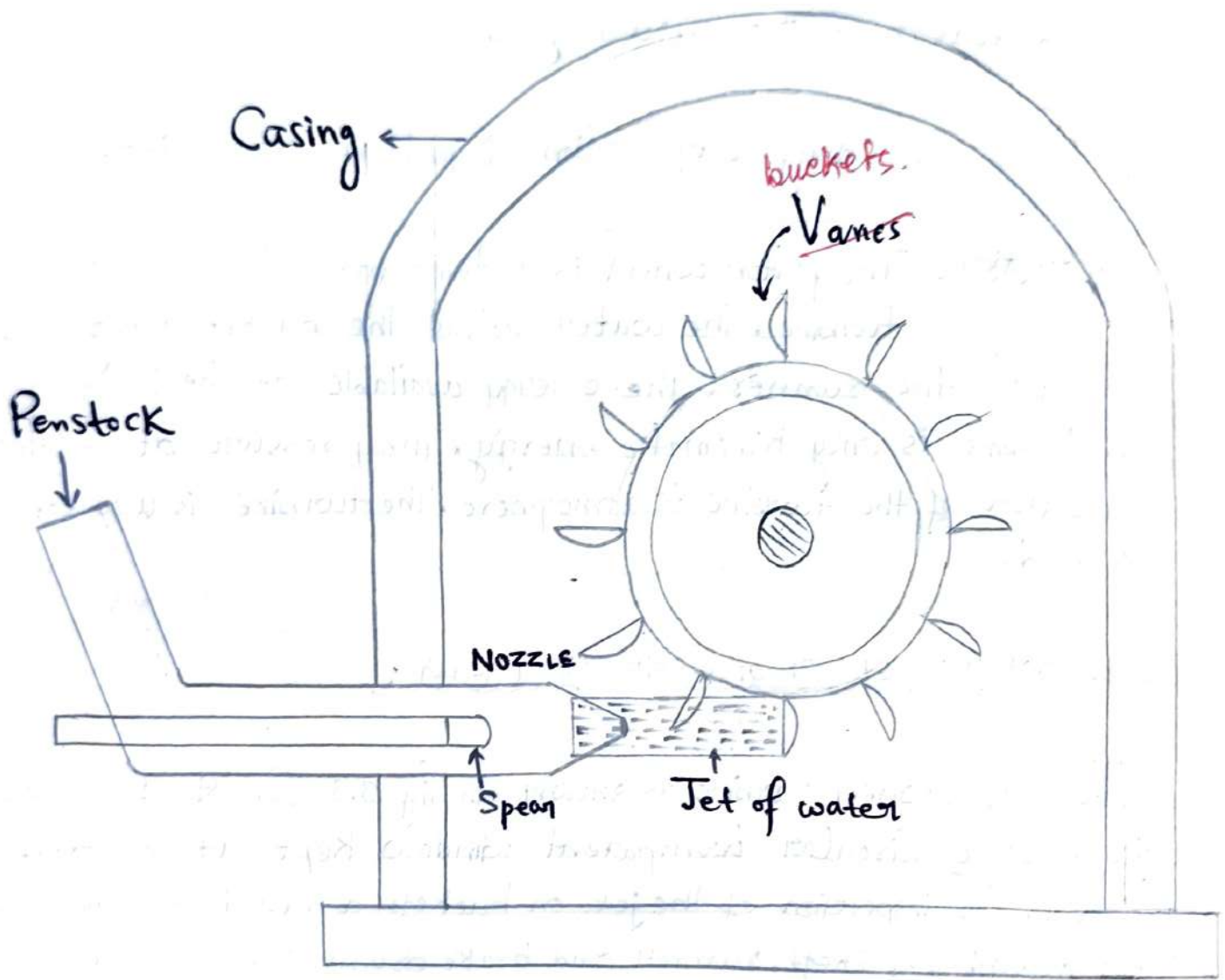


Fig:- pelton wheel Turbine line diagram.

The water coming out of the nozzle is formed into a free jet which impinges on a series of bucket of the runner thus causing it to revolve. The runner revolves freely in air the water is in contact with only a part of the runner at a time, and throughout its action on the runner.

Table -2: Specification of pelton wheel turbine.

- | | |
|-------------------|--|
| • Type | Centrifugal high speed, single Suction Volute. |
| • Power required | AC. 5HP/33 kW 3 phase, 440 Volts. |
| • Speed | 2880 rpm. |
| • Runner diameter | 210 mm. |
| • No. of buckets | 19 no.s. |
| • Rated speed | 1000 rpm. |

Experimental Procedure :

Starting Procedure :

- Cleaning of the apparatus is made and the tank is freed from dust.
- The drain valve provided is Closed.
- The Sump is filled to $\frac{3}{4}$ th with clean water with clean water and it is ensured that no foreign particles are present.
- The main power supply (440 V AC. 50 Hz) is switched ON.
- The pump is switched on with the help of starter.
- With 2 kg load as dead weights which add upto the load shown in the Spring balance (live weights) the following procedure is adopted.
 - (a) The gate of the guide vanes is open to 100%.
 - (b) The pressure readings on the two Venturimeters as well as the Vacuum gauge reading over the draught tube are noted down.
 - (c) The rpm of the shaft is noted down with the help of a tachometer.

① The gate of the guide vanes is opened to 50% with the help of hand wheel followed by point band C.

② The gate of the guide vanes is opened to 25% with the help of hand wheel followed by point b and C.

③ The gate of the guide vanes is now fully closed with the help of hand wheel followed by point's b and C.

Closing procedure :

- When an experiment with a particular load is over, the load is removed from dynamometer.
- The pump is switched OFF.
- After all the experiments are over, the main power supply is switched off.

Experimental Set-up (Schematic diagram) :

The layout of the pelton turbine in the hydraulic laboratory is sketched out in fig. 1. The parameters such as head and discharge calculation are formulated below.

There has not been a very significant difference in the values of discharge, as discharge is a function of differential pressure head which by itself has not shown much variation.

Comparatively the speed reduced with the increased in load. At 100% gate opening $N = 2473 - 1426$ rpm at 75%
gate opening $N = 2412 - 1405$ rpm at 50%
gate opening $N = 2400 - 1350$ rpm and at 25% of the
gate opening $N = 2270 - 1380$ rpm with spring balance
reading ranging from 2.7 - 7 kg while dead weight at 0.1 kg
throughout the experiment.

Total head and Discharge measurement the Set up

- 1.1 Head (H), Head is the change in the water levels between the hydro intake and the hydro discharge point. It is a vertical height measured in metres. The more the head, the higher the water pressure across the hydro turbine and the more power it will generate.

$$\text{Total head } W = 10 \times P \text{ (m of water)}$$

$$P = \text{Pressure gauge reading (kg/cm}^2\text{)}$$

1.2 Discharge (Q)

$$Q = A \times V \text{ m}^3/\text{sec}$$

$$\text{where } A = \text{Area of pipe m}^2 \\ = 0.00374 \text{ m}^2$$

$$V = C_v \sqrt{2gh} \times \left(\frac{\rho_m}{\rho_w} \approx 1 \right) \text{ m/sec}$$

$$[C_v = 0.98]$$

$$\rho_w = \text{Density of water, Kg/m}^3 = 1000 \text{ Kg/m}^3$$

$$\rho_m = \text{Density of manometer fluid ie Hg, Kg/m}^3 = 13600 \text{ Kg/m}^3$$

$$h = \text{Differential pressure of manometer m.}$$

1.3 Turbine input power (P_i)

$$\text{Turbine input, } P_i = \frac{\rho_w Q H}{75} \text{ HP}$$

$$= \frac{\rho_w Q H}{1000} \text{ KW}$$

ρ_w = density of water

1.4 Turbine output power (P)

$$P = \frac{2 g H N W R_e}{60} \text{ watt}$$

$$= \left[\frac{2 g H N W R_e}{60 \times 1000} \text{ KW} \right]$$

$$W = W_1 - W_2 \text{ Kg}$$

$$R_e = (D_b + 2 D_R) / 2 = 0.164 \text{ m}$$

$$D_b = 0.300 \text{ m}$$

$$D_R = 0.014 \text{ m}$$

$$g = \text{Acceleration due to gravity} = 9.81 \text{ m/s}^2$$

$$N = \text{Speed of the runner shaft (rpm)}$$

$$W_1 = \text{Dead weight (Kg)}$$

$$W_2 = \text{Spring balance weight}$$

$$R_e = \text{Effective radius} = 0.164 \text{ m}$$

Efficiency of turbine

$$\eta_{\text{turbine}} \% = \frac{P}{P_i} \times 100 \%$$

Data Collected at different loads and gate openings:

Expt no.	Load (kg)	Spring balance weight (kg)	Gate opening (%)	Pr gauge reading P (kg/cm ²)	Differential pressure h (m)	H (m)	Q m ³ /sec	N in rpm
	(W ₁)	(kg) W ₂			g g kg/cm ²			
1.	0.2	2.5	100	0.95	0.7	9.5	0.048	1370
2.	0.2	2.5	75	0.95	0.7	9.5	0.048	1374
3.	0.2	2.5	50	0.92	0.68	9.2	0.047	1368
4.	0.2	2.5	25	0.92	0.68	9.2	0.047	1385
5.	0.3	2.9	100	0.88	0.62	8.8	0.045	1280
6.	0.3	2.9	75	0.86	0.60	8.6	0.044	1306
7.	0.3	3.0	50	0.82	0.59	8.2	0.044	1306
8.	0.3	3.0	25	0.82	0.59	8.2	0.044	1300

Calculation Table:

Sl No.	Gate opening (%)	Head (H)	Speed N in rpm	Discharge Q (m ³ /sec)	Output (kN) P	Input (kN)	Turbine efficiency (%)
1.	100	9.5	1370	0.048	0.530	4.473	11.84
2.		8.8	1280	0.045	0.560	3.884	14.41
3.		9.5	1374	0.048	0.532	4.473	11.89
4.	75	8.6	1306	0.044	0.571	3.712	15.38
5.		9.2	1368	0.047	0.529	4.241	12.47
6.	50	8.2	1320	0.044	0.600	3.539	16.95
7.		9.2	1385	0.047	0.536	4.241	12.63
8.	25	8.2	1300	0.044	0.591	3.539	16.69

Result:

Efficiency of Turbine

$$\eta_{\text{turbine}} (\%) = \frac{P}{P_i} \times 100\%$$

$$= 17.71\%$$

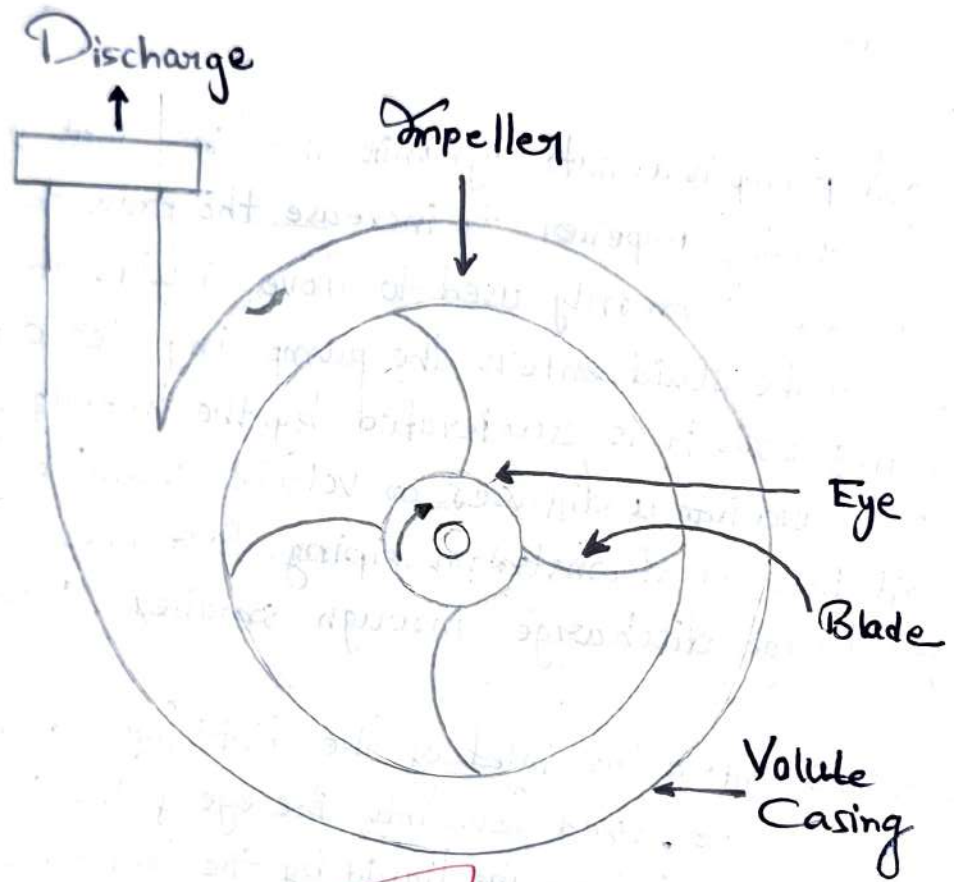
Aim: To Conduct the performance test on Centrifugal pump and to plot the operating Characteristics.

Introduction:

Centrifugal pump is a rate-dynamic machine that imparts energy to fluid by rotating impeller to increase the pressure of a fluid. These pumps are commonly used to move liquids through a piping system. When the fluid enters the pump impeller along or near to the rotating axis it is accelerated by the impeller, flowing radially outward into a diffuser or volute chamber (casing) from where it exits into the downstream piping system. Centrifugal pumps are used for large discharge through smaller heads.

The fluid enters the inlet of the Centrifugal pump under atmospheric pressure, and flows into the eye of the impeller. The Centrifugal force exerted on the liquid by the rotating impeller, moves the liquid away from the impeller eye and out along the impeller vanes to their extreme tip where the liquid is then forced against the inside wall of the volute and out through the discharge of the pump.

The energy is transferred from the shaft to the impeller and from the impeller to the water. There are three types of impeller based on the number of shrouds.



Good

Fig:- Centrifugal pump.

- Open Impeller - The vane are cast free on both side.
- Semi-open Impeller - The Vanes are free on one side and enclosed on the other.
- Closed Impeller - The vanes are located between two discs, all in a single casting.

Volute case: The Volute is a curve-shaped region that expands in cross-sectional area as it wraps around the pump casing. The purpose of the Volute is to collect the liquid discharged from the periphery of the impeller at high velocity and gradually cause a reduction in fluid velocity by increase the flow area. The Converts the Velocity head to static pressure. The fluid is then discharge from the Centrifugal pump through the discharge connection.

The performance characteristics of Centrifugal pump Can be determined by following formula.

Total - 01 :-

Initial reading (water level till crest) = 1.24 cm

Final reading = 15.26 cm

Head of water (H) = 14.02 cm

Actual discharge (Q_{act}) = 10428.9 cm^3/sec

$$\text{Delivery Head (H}_d\text{)} = 0.1 \text{ kg/cm}^2$$

$$\text{Delivery Head (H}_d\text{)} = 0.48 \text{ kg/cm}^2$$

$$\text{Suction Head (H}_s\text{)} = 4.8 \text{ m}$$

$$\text{Suction Head (H}_s\text{)} = 43 \text{ cm}$$

$$\begin{aligned}\text{Total head (H)} &= H_d + H_s + z \\ &= 1 + 4.8 + 43 \\ &= 6.23 \text{ m}\end{aligned}$$

$$\text{Wattmeter Reading (W}_p\text{)} = 425 \text{ W}$$

$$\text{Actual Discharge (Q}_{act}\text{)} = 10428.9 \text{ cm}^3/\text{sec}$$

$$\text{Total Head (H)} = 6.23 \text{ m}$$

$$\text{Input power} = 2040 \text{ W}$$

$$\text{Output power} = 637.4 \text{ W}$$

$$\begin{aligned}\text{Efficiency } \eta &= \frac{637.4}{2040} \times 100\% \\ &= 31.20\%\end{aligned}$$

Trial - 02 :-

$$\text{Initial reading (water level till crest)} = 1.24 \text{ cm}$$

$$\text{Final reading} = 14.95 \text{ cm}$$

$$\text{Head of water (h)} = 13.71 \text{ cm}$$

$$\text{Actual discharge (Q}_{act}\text{)} = 9861.96 \text{ m}^3/\text{sec}$$

$$\text{Delivery head (H}_d\text{)} = 0.2 \text{ kg/cm}^2$$

$$\text{Delivery head (H}_d\text{)} = 2 \text{ m}$$

$$\text{Suction Head (H}_s\text{)} = 0.46 \text{ kg/cm}^2$$

$$\text{Suction Head (H}_s\text{)} = 4.6 \text{ m}$$

$$z = 43 \text{ cm}$$

$$\text{Total head (H)} = 7.03 \text{ m}$$

$$\text{Wattmeter reading (W}_p\text{)} = 400 \text{ W}$$

$$\text{Input power} = 1920 \text{ W}$$

$$\text{Output power} = 680.12 \text{ W}$$

$$\text{Efficiency, } \eta = \frac{680.12}{1920} \times 100\% = 35.43\%$$

Trial - 03 :-

$$\text{Initial reading (water level till crest)} = 1.24 \text{ cm}$$

$$\text{Final reading} = 14.65 \text{ cm}$$

$$\text{Head of water (h)} = 13.41 \text{ cm}$$

$$\text{Actual discharge (Q}_{act}\text{)} = 9331.3 \text{ cm}^3/\text{sec}$$

$$\text{Delivery head (H}_d\text{)} = 0.3 \text{ kg/cm}^2$$

$$\text{Delivery head (H}_d\text{)} = 3 \text{ m}$$

$$\text{Suction Head (H}_s\text{)} = 0.42 \text{ kg/cm}^2$$

$$\text{Suction Head (H}_s\text{)} = 4.2 \text{ m}, z = 43 \text{ cm}$$

$$\text{Total Head (H)} = 3 + 4.2 + 43 = 7.63 \text{ m}$$

$$\text{Wattmeter reading (W}_R\text{)} = 400 \text{ W}$$

$$\text{Input power} = 1920 \text{ W}$$

$$\text{Output power} = 698.5 \text{ W}$$

$$\text{Efficiency, } \eta = \frac{698.5}{1920} \times 100\% = 36.38\%$$

$$\text{Average Efficiency } \eta = \frac{35.43 + 31.25 + 36.38}{3} = 34.35\%$$

Result :- The average efficiency of Centrifugal pump is 34.35%.

Kaplan Turbine Test Setup

Aim: • To determine the output power of Kaplan Turbine.

Objective: To Study the operation of a Kaplan turbine.

- To determine the efficiency of the Kaplan turbine
- To plot the performance characteristics Curve.

Introduction:

The Kaplan turbine is a reaction turbine like Francis turbine. It operates in an entirely closed conduit from inlet to race runner has two major difference firstly, in Francis runner. The water enters radially while in Kaplan type, water strikes on the blades axially. Secondly, the number of blades in Francis runner is only 16 to 24 but in the Kaplan. It is only 3 to 6. This reduces the contact surface with water and hence the frictional resistance. It is used for comparatively low head & large quantity of water is available.

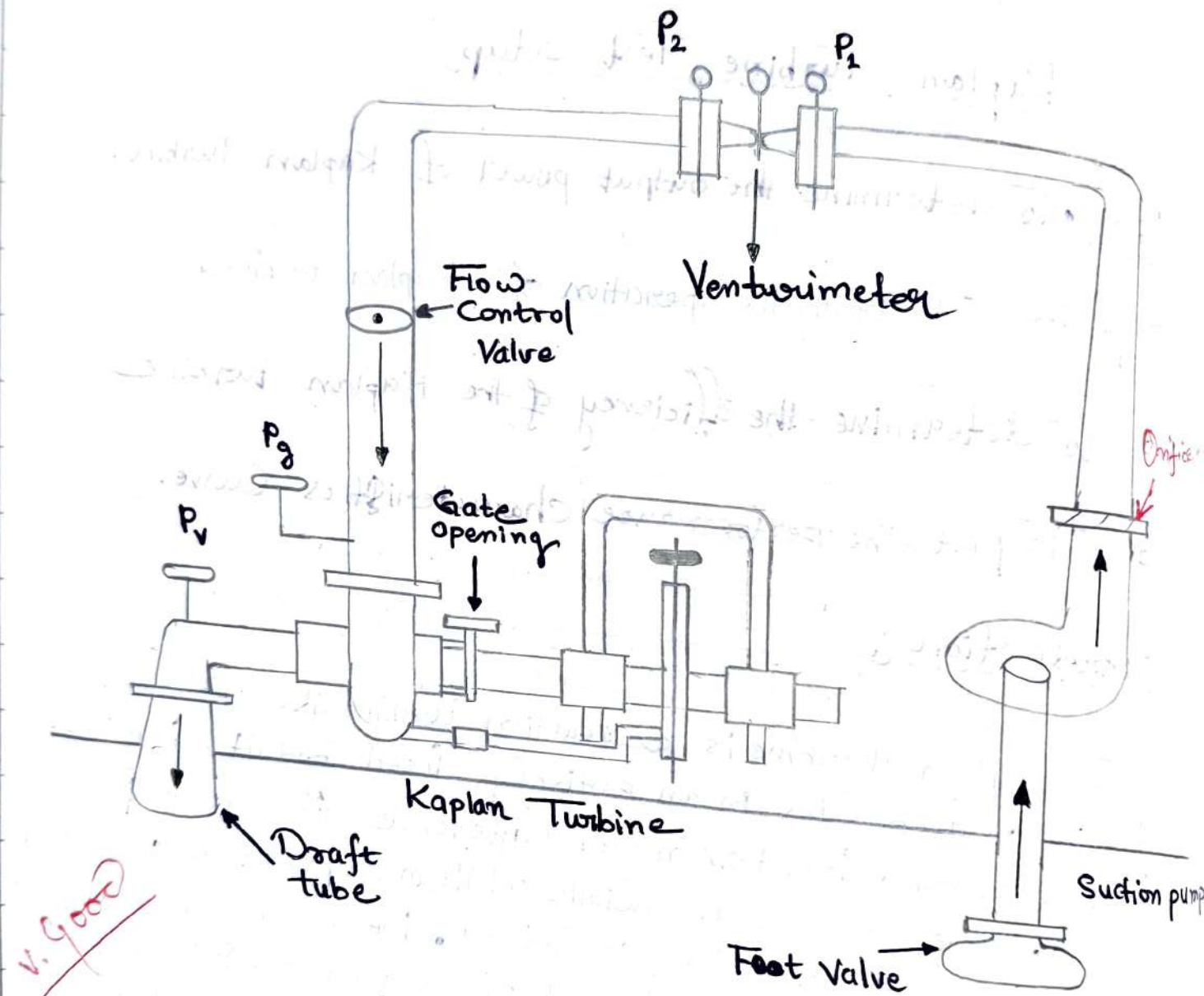


Fig. 1: Kaplan Turbine Set up.

Francis Turbine test Rig:

The Francis turbine as shown in fig-01 consists of spiral casing, an outer bearing pedestal & motor assembly with runners shaft & brake drum, all mounted on a suitable sturdy cast iron base plate. A straight conical draft tube is provided for the purpose of regaining the kinetic energy from the exit water & also facilitating easy accessibility of the turbine due to its locating at higher level than the tail race. A transparent hollow perspex cylinder is provided in between the draught bend and the casing for the purpose of observation of flow at exit of runner. A rope brake arrangement is provided to load the turbine. Adjusting the guide vanes for which a hand wheel and a suitable link mechanism are provided can control the output of the turbine. The net supply head on the turbine is measured by a pressure and vacuum gauge.

Specification of Francis formula:

Table 3.1 holds the specification of Francis turbine.

Table -1 (Specification of Francis turbine unit) Pump

Type	Centrifugal high speed single suction volute
* Power reqd.	A.C. 5 HP/ 3.7 kW 3 phase, 440 volts
* Dial gauge	10 kg
* Runner diameter	162 mm
* No-s of guide Vanes	10 no.s
* Head	8 - 22.6 metre
* Speed	2870 - 3100 rpm

EXPERIMENTAL PROCEDUREStarting Procedure:

- (i) Cleaning of the apparatus is made & the tank is forced free from dust.
- (ii) The drain valve provided is closed.
- (iii) The sump is filled to $\frac{3}{4}$ th with clean water & it is ensured that no free & particle are present.
- (iv) The main power supply (440 V A.C. 50 Hz) is switch ON.
- (v) The pump is Switched on with the help of starter.
- (vi) With 2kg load as dead weight which add up to the load shown in the opening balance (live weight) the following procedure is adopted -

- (a) The gate of the guide vanes is opened to 100%.
- (b) The pressure readings on the two Venturimeter, as well as the vacuum gauge reading over the draught tube are noted down.
- (c) The rpm of the shaft is noted down with the help of a tachometer.
- (d) The gate of the guide vanes is opened to 75% with the help of hand wheel followed by b & c.
- (e) The gate of the guide is opened to 50% with the help of hand wheel followed by b & c.
- (f) The gate of the guide vanes is opened to 25% with the help of hand wheel by point b & c.
- (g) The gate of the guide vanes is now fully closed with the help of hand wheel followed by point b & c.

With 3.5 & 6 kg as dead weight, procedure a to g is followed.

Closing Procedure :

* When an experiment with a particular load is over, the load is removed from dynamometer.

* The pump is switched off.

* After all the experiment are over the main power supply is switched off.

3. Experimental set-up (schematic diagram)

Total head and discharge measurement is set-up.

4.1 Head (H) :

Head is the change in water levels between the hydro intake & the hydro discharge point. It is a vertical height measured in metres. The more the head the higher the water pressure across the hydro turbine and the more power it will generate.

Result / conclusion :-

Aim: Determination of the Co-efficient of friction for pipes.

Objectives: To determine the Co-efficient of friction for pipe of different size.

Apparatus Used:- pipe friction apparatus.

Introduction:

Transportation of fluids through pipes is frequently deal with by engineers. Distribution of water and gas for domestic consumption through pipes is an example. Experimental observation by Froude on long, straight & uniform diameter pipes on the flow of water indicates that head loss due to friction h_f between two section of the pipes varied in direct proportion with the velocity head V^2 the distance between the section 1 and 2 l inversely with pipe diameter (d) by introducing a Co-efficient of proportionality (f) and called the friction factor. Darcy weisbach proposed the following equation for head loss due to friction a pipe $h_f = f \frac{L}{D} \times \frac{V^2}{2g}$

Description of Apparatus:

The apparatus consists of a hydraulic bench where a reservoir tank and a collecting tank are attached. A motor of 0.5 HP is used to circulate the water within the arrangement is arranged within the

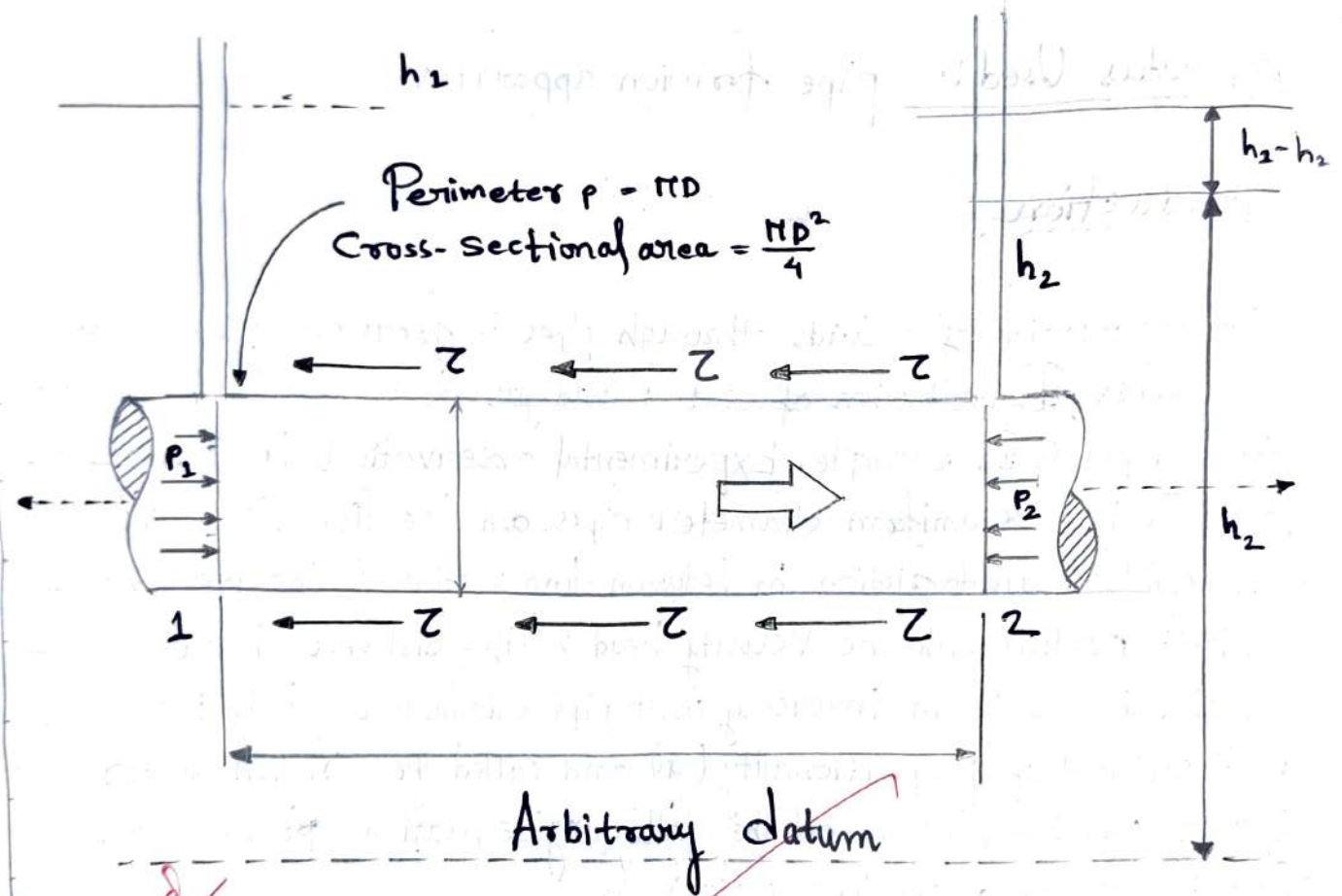


Fig:- Friction (major) losses in pipe.

flow path to measure the pressure difference from which we can measure the head losses. The diameter of the flexible tubes attached to the manometer are 1.5 cm & 2 cm respectively.

Experimental Procedure :

- (i) open the air vessel & release the air on manometric tube & weight the levelling of mercury, close the air vessel valve.
- (ii) Switch on the power and start the motor.
- (iii) Record the diameter ' d ' of the pipe and the length ' L ' between the sections attached to the limbs of u-tube manometer.
- (iv) Open the supply valve to allow water to flow in one pipe only.
- (v) Record the initial water ' y_1 ' in the piezometer fitted to the discharge measuring tank and start the stop watch & find the depth of water collected for a particular time by reading the final reading ' y_2 ' of the piezometer.
- (vi) Knowing the area of measuring tank flow discharge through the pipe can be obtained.
- (vii) Record the readings of the two limbs of the inverted U-tube manometer the difference of which gives the head loss h_1 .
- (viii) Calculate average velocity through the pipe using the relationship $V = \frac{Q}{a}$ where,
 $a = \text{Area of Cross-Section of the pipe.}$

(ix) The Darcy weisbach Co-efficient of friction is Calculated using the expression.

$$h_1 = f \frac{L}{D} \times \frac{V^2}{2g} \text{ (Pipe friction)}$$

$$f = \frac{2g D h_1}{L V^2}$$

Where h_1 is the head loss 'f' is the friction factor 'L' is the pipe length 'V' is the avg. flow Velocity & g is the acceleration due to gravity.

(x) Repeat the above steps for different discharge and other pipe of different diameter taking at least 3 reading for every pipe.

Aim: To Conduct a performance test on double acting reciprocating pump and to plot the operating characteristics.

Introduction:

The reciprocating pump is a positive displacement pump, it operates on the principle of actual displacement or pushing of liquid by a piston or plunger that executes a reciprocating motion in a closely fitted cylinder. The liquid is alternatively drawn from the pump and filled into suction side of the cylinder. The liquid fed to discharge side of the cylinder and emptied to the delivery pipe. There are three types of reciprocating pumps.

- (i) plunger pumps
- (ii) Diaphragm pumps
- (iii) piston pumps

(i) plunger pumps: In this type of pumps, the cylinder is fitted with a plunger. This plunger creates the pressure on the water and discharges it through the discharge valve.

(ii) Diaphragm pumps: The plunger pressurizes hydraulic oil which is used to flex a diaphragm in the

Types of Reciprocating Pumps:

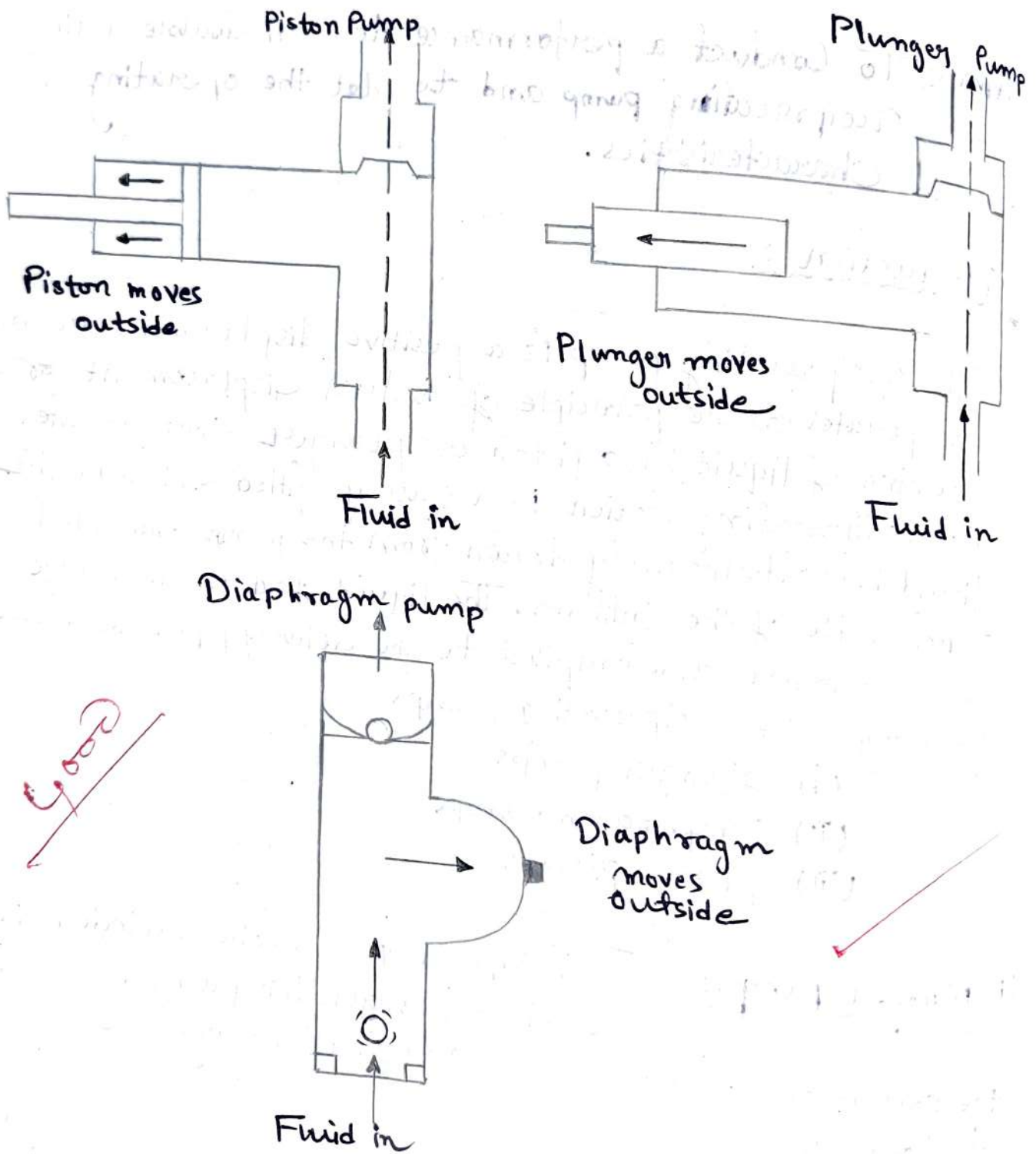


Fig:- When all pump at suction.

Pumping Cylinder. Diaphragm valves are used to pump hazardous and toxic fluids.

(iii) Piston pumps: In this type of reciprocating pump piston moves and pressurizes the fluid by using one or more reciprocating pistons. It is typically driven by an electric motor through Crankshaft and Connections rod.

According to the Contact between the plunger and water reciprocating pump is Classified as Single acting reciprocating pump and double acting reciprocating pump. A Single acting pump is one which has one Suction valve, delivery valve and one Suction and delivery pipe.

Components of reciprocating pump:

Crank, Connecting rod, piston, cylinder, Suction pipe, Suction valve, Delivery pipe, Strainer, Air Vessel, working of air vessel.

Purpose of Air vessel:

- (i) To obtain liquid at uniform discharge.
- (ii) Due to air vessel frictional head and acceleration head decreases and the work overcoming friction resistance in Suction and delivery pipe Considerably decreases which results

in good amount of work.

- (iii) Reduces the possibility of Separation and Cavitation.
- (iv) Reciprocating pump can run at high speed without flow Separation.
- (v) Suction pipe can be increased by increasing the length of pipe below air vessel.
- (vi) Large amount of power is saved due to low acceleration head.

Comparison of Centrifugal and Reciprocating pump:

Centrifugal pump

- (i) Steady flow
- (ii) Uniform torque
- (iii) High efficiency for low head pump high discharge.
- (iv) Low initial cost
- (v) Low maintenance cost

Reciprocating pump

- (i) pulsating flow
- (ii) Torque is not uniform
- (iii) Low efficiency for low head pumps.
- (iv) High initial cost
- (v) Low maintenance cost

Let, D = Diameter of the Cylinder.

A = ~~Radius~~ area of the piston or cylinder.

r = Radius of Crank.

N = rpm of the tank.

L = Length of the stroke $2r$

Number of revolution per second = $\frac{N}{60}$

$$Q_{th} = \frac{A L N}{60} + \frac{(A - A_D) L N}{60}$$

$$Q_{th} = \frac{2 A L N}{60}$$

$$Q_{act} = C_d \times Q_{th}$$

$$Q_{act} = \frac{A \times h}{t}$$

$$\eta_{motor} = \frac{\text{Output motor}}{\text{Input motor}}$$

$$\eta_{pump} = \frac{\text{Output of pump}}{\text{Input of pump}}$$

$$\eta_{overall} = \frac{\text{Output of pump}}{\text{Input of pump}}$$

$$\text{Percentage Slip} = \frac{Q_{th} - Q_{act}}{Q_{th}}$$

PROCEDURE:

- (i) Open the double acting reciprocating pump and Click on the next button shown at the bottom right corner and also description provided.
- (ii) Click on the green button to start the pump and Click on next button.

- (iii) Click on the hand to rotate the inlet valve and allow the water to flow.
- (iv) Note the speed of motor as shown in the Circle and Click on next button.
- (v) Click on delivery head gauge to show the delivery head.
- (vi) Note the delivery head reading and Click on next button.
- (vii) Click on Suction head gauge to Show the Suction head.
- (viii) Note the Suction head reading and Calculate the total head, Click on next button.
- (ix) Click on the hand wheel to get spring balance reading.
- (x) Click on ~~10~~ Calculate the torque reading and click on next button.
- (xi) Click on to Close the stopper and note down the time required to fill 10 cm.
- (xii) Note down the time and Calculate the actual discharge.
- (xiii) Note the time taken by energy meter for 5 revolution, then click on next.
- (xiv) Calculate the efficiency of motor and pump using the formula, then Click on next button to continue.
- (xv) Click the red button to stop the pump then Click on next button.
- (xvi) Click on each label button to see the different graphs.

Calculation:

Specific weight of water (γ) = 9810 N/m^3

Time taken by energy meter for 5 revolution (t_2) = 43.5 sec

Actual discharge (Q_{act}) = $5.68 \times 10^{-4} \text{ m}^3/\text{sec}$

Total head (H) = 0.968 m

Speed of motor (N_m) = 1200 rpm

Speed of pump (N_p) = 240 rpm

Torque (τ) = 1.32 N·m

Motor input = 275.86 W

pump output = 5.39 W

Theoretical discharge, Q_{th} = $0.0007 \text{ m}^3/\text{sec}$

percentage slip = 25.27 %

Motor output = 165.88 W

Efficiency of motor = 60.12 %

Efficiency of pump = 3.25 %

Result:

overall efficiency,

$$\eta = 1.95 \%$$

Title Study of hydraulic force.

Objectives:- To study the effort of force on following types of Vanes:- Hemispherical vanes and flat vanes.

Apparatus:- Impact of jet apparatus.

Introduction:- When a plate is placed in the path of a jet, the jet exerts a force on the plate. This force can be calculated from the impeller momentum equation. Momentum equation is based on Newton's 2nd law of motion, which states that the "The algebraic sum of external force applied to control volume of fluid in any direction is equal to the rate of change of momentum in that direction."

THEORY:-

An a vertical water jet moving with Velocity 'V' is made to strike a target which is free to move in the vertical direction, then a force will be exerted at the target by impact of jet. According to momentum equation this force must be equal to the rate of change of momentum of the jet flow in the same direction. Due to the impact of the jet on the flat stationary plate the entire velocity of the jet is destroyed. But the plate is at right angles to the jet. Hence the components of the velocity of the jet in the direction of the jet after striking it will be zero.

$F \rightarrow$ Rate of Change of momentum in the direction of force.

$$F = \frac{\text{Initial momentum} - \text{Final momentum}}{\text{Time}}$$

$$= \frac{\text{mass} \times \text{initial Velocity} - \text{mass} \times \text{final Velocity}}{\text{Time}}$$

$$= \frac{\text{mass}}{\text{Time}} \times (\text{Initial Velocity} - \text{Final Velocity})$$

$$= \rho A V (V - 0)$$

$$= \rho A V^2 \longrightarrow \text{For flat Vane}$$

$$F_u = 2 \rho A V^2 \longrightarrow \text{For hemispherical Vane}$$

Description:

The Setup Consist of a Sump tank with Centrifugal pump to Circulate water. A chamber with side glass is provided for Visualization of impact of jet on Vanes. Water from Sump tank flows through a nozzle and Strikes Vertically to Vane positioned above the nozzle.

Utilities Required:

- $\rightarrow$ Electricity Supply : Single phase, 220 AC, 50 Hz 5-15 amp. Combined Socket.
- $\rightarrow$ Floor drain required.
- $\rightarrow$ Water Supply.
- $\rightarrow$ Floor area required 1.5 m X 0.1 m.

EXPERIMENTAL PROCEDURE:

→ Starting procedure:

1. Close all the Vanes provided V_1 to V_4 .
2. Fill sump tank $\frac{3}{4}$ with water and ensure that foreign particles are there.
3. Fix desired Vane (Hemispherical vane).
4. Ensure that all ON/OFF Switches given on the panel are at OFF position.
5. Open the pan Valve, V_2 .
6. Switch ON the pump.
7. put weight on the pan.
8. Operate by pass Valve V_2 and flow control Valve.
9. Now Control the flow of water so that the applied weight on the top is Counter balanced by the impact of jet.
10. Measure flow rate using measuring tank and stop watch.
11. Repeat the experiment for different weights.
12. Repeat the experiment for other Vane.

→ Closing procedure:

1. When experiment is over, Switch off the pump.
2. Switch off power supply to panel.
3. Drain water from all tanks with the help of drain Valves V_4 & V_3 .

Observation Table

We Consider only hemispherical vane

Length of channel = 60 cm

Breadth of channel = 60 cm

Area of Collecting tank = 3600 cm² (A)

Diameter of nozzle, $d = 12.5$ cm

Area of the nozzle, $A = \frac{\pi}{4} \times 12.5^2$
 $= 122.66$ cm²

Discharge of water = 418.6 cm³/sec

Density of water, $\gamma = 1$ g/cm³.

Calculation :Trial-1 :

$$a = 13.55 \text{ cm}$$

$$b = 63 \text{ cm}$$

Weight on the pan, $w = 50$ gm

Time taken by water to fill, $t = 43$ Sec

Height, $h = 5$ cm

$$\therefore \text{Discharge, } Q = \frac{A \times h}{t} = \frac{3600 \times 5}{43}$$

$$= 418.6 \text{ cm}^3/\text{Sec}$$

Now, Velocity of jet, $v = Q/A$

$$= 337.88 \text{ cm/Sec}$$

$$\begin{aligned}\text{Theoretical force, } F_{th} &= 2\gamma AV^2 \\ &= 2 \times 1 \times 3600 \times 337.88^2 \\ &= 2.83 \text{ N}\end{aligned}$$

$$\begin{aligned}\text{Actual force} &= \frac{Wxb}{13.55} \\ &= 2.29 \text{ N}\end{aligned}$$

$$\begin{aligned}\therefore \text{Co-efficient of impact, } C_i &= \frac{F_{act}}{F_{th}} \\ &= 2.83 / 2.29 = 0.81\end{aligned}$$

Trial-2:-

$$a =$$

$$b =$$

$$\begin{aligned}\text{Weight on the pan, } W &= 100 \text{ gm} \\ t &= 33.5 \text{ Sec}\end{aligned}$$

$$h = 5 \text{ cm}$$

$$\begin{aligned}\therefore \text{Discharge, } Q &= \frac{Axh}{t} = \frac{3600 \times 5}{33.5} \\ &= 537.31 \text{ cm}^3/\text{sec}\end{aligned}$$

Now,

$$\begin{aligned}\text{Velocity of jet, } V &= Q/A = 433.7 \text{ cm/sec} \\ F_{th} &= 2\gamma AV^2 \\ &= 2 \times 1 \times 3600 \times 433.7^2 \\ &= 9.66 \text{ N}\end{aligned}$$

$$F_{act} = \frac{Wxb}{a} = 4.578 \text{ N}$$

$$\therefore \text{Co-efficient of impact, } C_i = \frac{F_{act}}{F_{th}} = 0.98$$

Trial-3:weight on the pan, $w = 150 \text{ gm}$

$$t = 27 \text{ Sec}$$

$$h = 5 \text{ cm}$$

$$\therefore \text{Discharge, } Q = \frac{A \times h}{t} = \frac{3600 \times 5}{27}$$

$$= 666.67 \text{ cm}^3/\text{sec}$$

Now.

Velocity of the jet, $V = Q/A = 538.11 \text{ cm/sec}$

$$F_{th} = 2g A V^2$$

$$= 2 \times 1 \times 3600 \times 538.11^2$$

$$= 7.17 \text{ N}$$

$$F_{act} = \frac{W \times b}{a}$$

$$= 6.867 \text{ N}$$

$$\therefore \text{Co-efficient of impact, } C_i = \frac{F_{act}}{F_{th}}$$

$$= 0.96$$

$$\text{The average co-efficient of impact} = \frac{0.81 + 0.98 + 0.96}{3}$$

$$= 0.916$$

Result: The average co-efficient of impact of jet is 0.916.